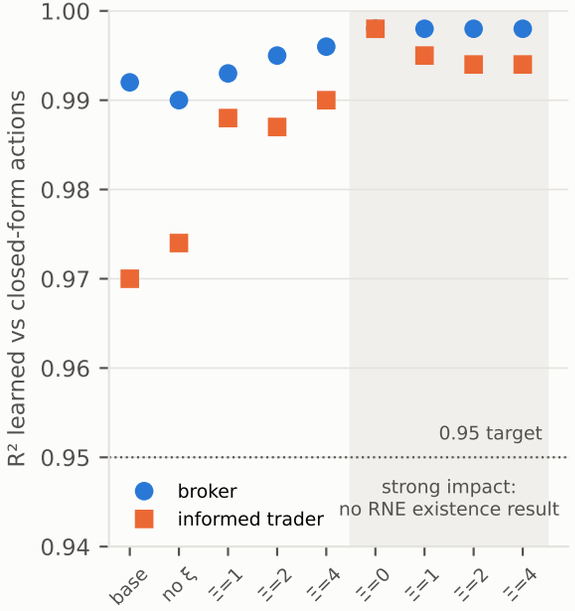
Learned broker strategies against informed order flow

Benchmark: Riccati–Nash equilibrium (RNE) of Cartea, Jaimungal & Sánchez-Betancourt (2026) | certified multi-agent learning → 4.3 M Binance BTC/USDT trades → 11 RL families, 144 held-out hours

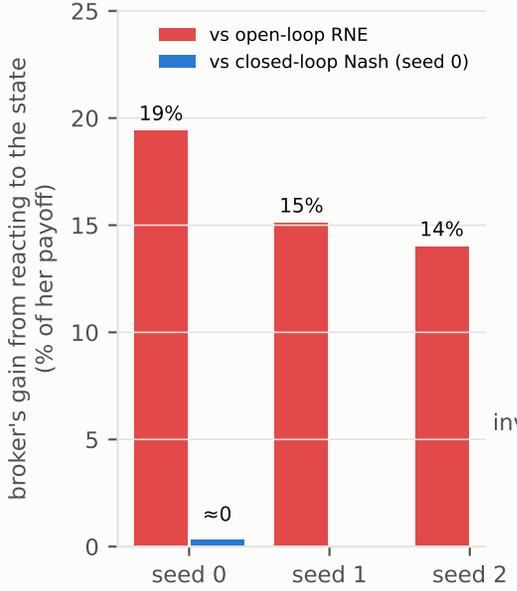
① Equilibrium recovered

$R^2 \geq 0.97$ in all 9 RNE configurations;
exploitability < 1% of payoffs



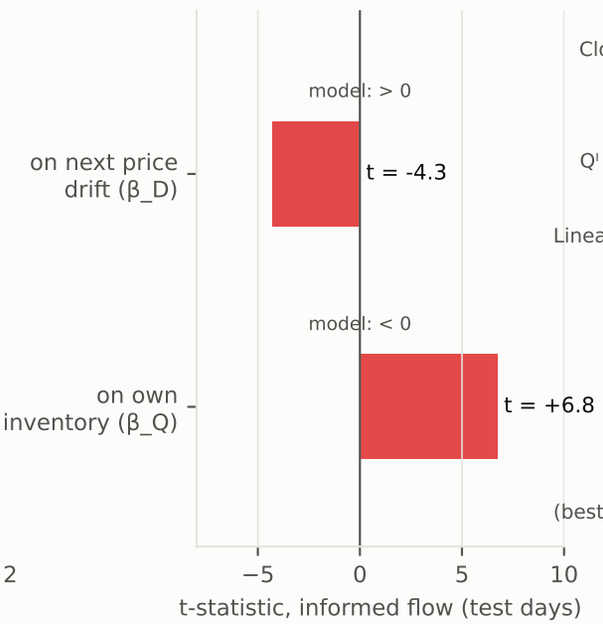
② Open loop is exploitable

against the RNE, a broker
who reacts to the state gains 14–20%



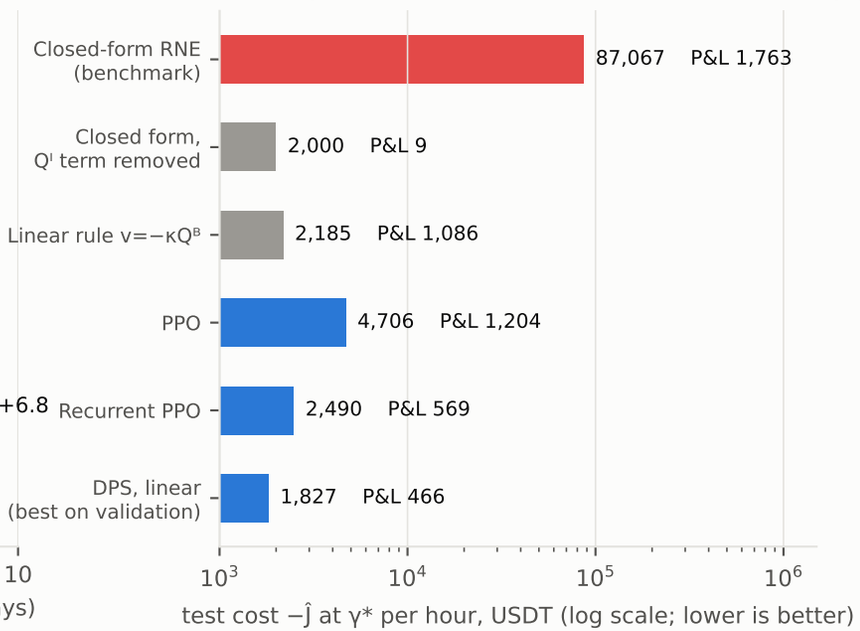
③ Model assumptions fail

large orders reverse and persist,
opposite to the model; signal $R^2 < 0$



④ Out of sample, paper's own objective

8 of 11 RL families beat the closed form (Holm p<0.01);
removing one term of the closed form closes most of the gap



Takeaway: the closed form is a sound equilibrium computation at its own parameters, but its response to the informed client's inventory fails out of sample.
Removing that term, or learning the controller, cuts the risk-adjusted cost about 40×; raw P&L still favours the original closed form.